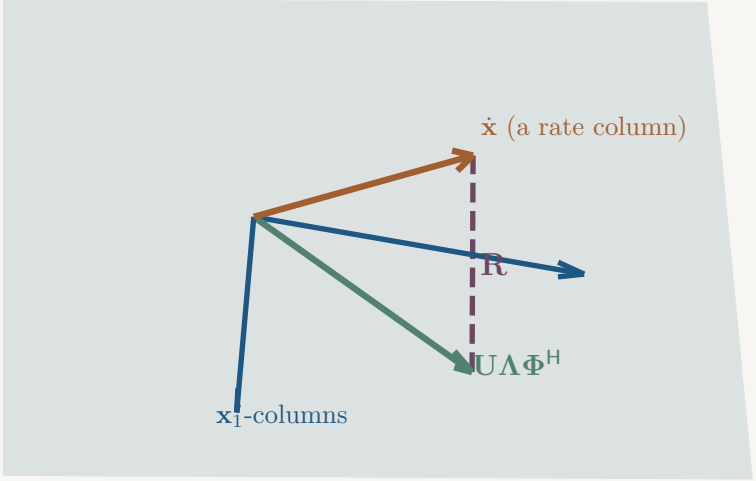


$$\mathbf{X}_1 = \mathbf{U}\boldsymbol{\Phi}^{\mathsf{H}} \quad \text{exact}$$

$$\mathbf{X}_2 = \mathbf{U}\boldsymbol{\Lambda}\boldsymbol{\Phi}^{\mathsf{H}} + \mathbf{R}$$

$$\mathbf{R} = (\mathbf{I} - \mathbf{X}_1\mathbf{X}_1^{\dagger})\mathbf{X}_2 \;\perp\; \operatorname{span}\mathbf{X}_1$$

linear, noise-free: $\mathbf{R} = \mathbf{0}$

nonlinear window: $\|\mathbf{R}\|/\|\mathbf{X}_2\|$

is what no linear model of the window explains